Reg. No.: ……

Name: ……

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU–UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 Admission Onwards)

MG4DSCSTA203 – APPLIED STATISTICAL METHODS

Duration: 1½ Hours Maximum Marks: 50

PART A – Short Answer Type Questions

Answer any 7 out of 10 questions. Each question carries 2 marks.

1. Why is Fisher's index number called an ideal index number? [U][CO1]

2. Define an index number. [R][CO1]

3. What is a time series? [U][CO2]

4. Define seasonal variation. [U][CO2]

5. What is statistical inference? [U][CO3]

6. What do you mean by a two-tailed test? [U][CO3]

7. Define the factor reversal test. [U][CO1]

8. Write the additive model of a time series. [U][CO2]

9. State any two assumptions of parametric tests. [R][CO3]

10. What is deseasonalisation? [U][CO2]

(7 × 2 = 14 marks)

PART B – Short Essay Type Questions

Answer any 4 out of 6 questions. Each question carries 6 marks.

11. Explain the uses and limitations of index numbers. [U][CO1]

12. Obtain the trend values by the semi-average method for the following data. [A][CO2]

Year:	2020	2021	2022	2023	2024	2025
Value:	110	105	115	110	120	130

13. Define the Time Reversal Test. Examine whether Laspeyres' and Fisher's index numbers satisfy this test. [An][CO1]

14. Explain the importance of time series analysis. [U][CO2]

15. Distinguish between Type I error and Type II error in testing of hypothesis. [An][CO3]

16. The records of a certain hospital showed the birth of 723 males and 617 females in a certain week. Do these data conform to the hypothesis that the sexes are born in equal proportions? (Level of significance = 0.05) [A][CO3]

(4 × 6 = 24 marks)

PART C – Essay Type Questions

Answer any 1 out of 2 questions. Each question carries 12 marks.

17. Construct Laspeyres', Paasche's and Fisher's index numbers for the following data. [A][CO1]

Items	P_0	Q_0	P_1	Q_1
A	9	120	16	150
B	12	90	14	100
C	8	60	12	56
D	6	40	10	30

18. A sample of 200 boys who passed the S.S.L.C. examination has a mean mark of 50 with a standard deviation of 5 in English. The mean mark of 100 girls is 48 with a standard deviation of 4. Does this indicate any significant difference between the abilities of boys and girls, assuming equal population standard deviations? (Level of significance = 0.05) [A][CO3]

(1 × 12 = 12 marks)